第五节 多元函数的泰勒公式

一、问题的提出

一元函数的泰勒公式：

$$f(x)=f(x_0)+f'(x_0)(x-x_0)$$
$$+\frac{f''(x_0)}{2}(x-x_0)^2+\cdots+\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$
$$+\frac{f^{(n+1)}\big(x_0+\theta(x-x_0)\big)}{(n+1)!}(x-x_0)^{n+1}\qquad(0<\theta<1).$$

意义：可用n次多项式来近似表达函数$f(x)$，且误差是当$x\to x_0$时比$(x-x_0)^n$高阶的无穷小.

二、中值定理和泰勒公式

定理 8 设$z=f(x,y)$在凸域$D\subset R^2$上连续, 在D的所有内点可微，则对D内任意两点$P(a,b)$, $Q(a+h,b+k)\in \text{int}\,D$, $s.t$

$$f(a+h,b+h)-f(a,b)$$
$$=f'_x(a+\theta h,b+\theta k)h+f'_y(a+\theta h,b+\theta k)k,\ (0<\theta<1)$$

证 引入函数 $\Phi(t)=f(a+ht,b+kt)$, $(0\le t\le 1)$.

显然 $\Phi(t)\in C[0,1]$, 在$(0,1)$内可微 ,由中值定理

$$f(a+h,b+k)-f(h,k)=\Phi(1)-\Phi(0)=\Phi'(\theta).$$
$$=f'_x(a+\theta h,b+\theta k)h+f'_y(a+\theta h,b+\theta k)k,$$
$$(0<\theta<1).$$

定理 9 设$z=f(x,y)$在点(x_0,y_0)的某一邻域内连续且有直到$n+1$阶的连续偏导数，(x_0+h,y_0+h)为此邻域内任一点, 则有

$$f(x_0+h,y_0+h)=f(x_0,y_0)+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(x_0,y_0)$$

$$+\frac{1}{2!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(x_0,y_0)+\cdots+\frac{1}{n!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(x_0,y_0)$$

$$+\frac{1}{(n+1)!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n+1} f(x_0+\theta h,y_0+\theta k),\qquad (0<\theta<1)$$

其中记号

$$\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(x_0,y_0)$$

表示 $hf'_x(x_0,y_0)+kf'_y(x_0,y_0)$,

$$\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(x_0,y_0)$$

表示 $h^2 f''_{xx}(x_0,y_0)+2hkf''_{xy}(x_0,y_0)+k^2 f''_{yy}(x_0,y_0)$,

一般地,记号 $\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^m f(x_0,y_0)$表示

$$\sum_{p=0}^{m} C_m^p h^p k^{m-p} \frac{\partial^m p}{\partial x^p \partial y^{m-p}}\bigg|_{(x_0,y_0)}.$$

证　　引入函数

$$\Phi(t)=f(x_0+ht,y_0+kt),\quad (0\le t\le 1).$$

显然 $\Phi(0)=f(x_0,y_0)$,

$$\Phi(1)=f(x_0+h,y_0+k).$$

由$\Phi(t)$的定义及多元复合函数的求导法则,可得

$$\Phi'(t)=hf'_x(x_0+ht,y_0+kt)+kf'_y(x_0+ht,y_0+kt)$$

$$=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(x_0+ht,y_0+kt),$$

$$\Phi''(t)=h^2f''_{xx}(x_0+ht,y_0+kt)$$

$$+2hkf''_{xy}(x_0+ht,y_0+kt)+k^2f''_{yy}(x_0+ht,y_0+kt)$$

…… ……

$$\Phi^{(n+1)}(t)=\sum_{p=0}^{n+1}C_{n+1}^{p}h^{p}k^{n+1-p}\frac{\partial^{n+1}p}{\partial x^{p}\partial y^{n+1-p}}\bigg|_{(x_0+ht,y_0+kt)}$$

$$=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n+1}f(x_0+ht,y_0+kt).$$

利用一元函数的麦克劳林公式，得

$$\Phi(1)=\Phi(0)+\Phi'(0)+\frac{1}{2!}\Phi''(0)+\cdots$$

$$+\frac{1}{n!}\Phi^{(n)}(0)+\frac{1}{(n+1)!}\Phi^{(n+1)}(\theta),(0<\theta<1).$$

将 $\Phi(0)=f(x_0,y_0)$，$\Phi(1)=f(x_0+h,y_0+k)$ 及上面求得的$\Phi(t)$直到n阶导数在$t=0$的值, 以及$\Phi^{(n+1)}(t)$在$t=\theta$的值代入上式. 即得

$$
\begin{aligned}
f(x_0+h,y_0+k) &= f(x_0,y_0)+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(x_0,y_0)\\
&+\frac{1}{2!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(x_0,y_0)+\cdots\\
&+\frac{1}{n!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(x_0,y_0)+R_n, \qquad (1)
\end{aligned}
$$

其中

$$R_n=\frac{1}{(n+1)!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n+1}f(x_0+\theta h,y_0+\theta k),$$

$$(0<\theta<1). \qquad (2)$$

证毕

公式(1)称为二元函数$f(x,y)$在点(x_0,y_0)的n阶泰勒公式，而R_n的表达式(2)称为拉格朗日型余项.

由二元函数的泰勒公式知， R_n 的绝对值在点(x_0, y_0)的某一邻域内都不超过某一正常数M.于是, 有下面的误差估计式:

$$|R_n| \le \frac{M}{(n+1)!}(|h|+|k|)^{n+1} = \frac{M}{(n+1)!}\rho^{n+1}(|\cos\alpha|+|\sin\alpha|)^{n+1}$$

$$= \frac{(\sqrt{2})^{n+1}}{(n+1)!}M\rho^{n+1}, \quad \text{其中 } \rho = \sqrt{h^2+k^2}. \qquad (3)$$

由(3)式可知, 误差$|R_n|$是当$\rho \to 0$时比ρ^n高阶的无穷小.

当$n=0$时, 公式(1)成为

$$
\begin{aligned}
& f(x_0+h, y_0+k) \\
& \quad = f(x_0, y_0)+hf'_x(x_0+\theta h, y_0+\theta k) \\
& \quad + kf'_y(x_0+\theta h, y_0+\theta k)
\end{aligned}
$$

上式称为二元函数的拉格朗日中值公式.

推论 如果函数$f(x,y)$的偏导数$f'_x(x,y)$, $f'_y(x,y)$在某一邻域内都恒等于零, 则函数$f(x,y)$在该区域内为一常数.

在泰勒公式(1)中，如果取 $x_0 = 0, y_0 = 0$，

则(1)式成为n阶麦克劳林公式.

$$f(x,y) = f(0,0) + \left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)f(0,0)$$

$$+\frac{1}{2!}\left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)^2 f(0,0) + \cdots + \frac{1}{n!}\left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)^n f(0,0)$$

$$+\frac{1}{(n+1)!}\left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)^{n+1} f(\theta x, \theta y),$$

$$(0 < \theta < 1) \qquad (5)$$

例 1 求函数 $f(x,y)=\ln(1+x+y)$ 的三阶麦克劳林公式.

解 $\because \quad f'_x(x,y)=f'_y(x,y)=\dfrac{1}{1+x+y}$,

$$f''_{xx}(x,y)=f''_{xy}(x,y)=f''_{yy}(x,y)=-\frac{1}{(1+x+y)^2},$$

$$\frac{\partial^3 f}{\partial x^p\partial y^{3-p}}=\frac{2!}{(1+x+y)^3}, \qquad (p=0,1,2,3),$$

$$\frac{\partial^4 f}{\partial x^p\partial y^{4-p}}=-\frac{3!}{(1+x+y)^4}, \qquad (p=0,1,2,3,4),$$

$$\therefore \left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)f(0,0)=xf'_x(0,0)+yf'_y(0,0)=x+y,$$

$$\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^2 f(0,0)$$

$$=x^2 f''_{xx}(0,0)+2xyf''_{xy}(0,0)+y^2 f''_{yy}(0,0)$$

$$=-(x+y)^2,$$

$$\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^3 f(0,0)=x^3 f'''_{xxx}(0,0)+3x^2 yf'''_{xxy}(0,0)$$

$$+3xy^2 f'''_{xyy}(0,0)+y^3 f'''_{yyy}(0,0)=2(x+y)^3,$$

又 $f(0,0)=0$, 故

$$\ln(1+x+y)=x+y-\frac{1}{2}(x+y)^2+\frac{1}{3}(x+y)^3+R_3,$$

其中

$$R_3=\frac{1}{4!}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^4 f(\theta x,\theta y)$$

$$=-\frac{1}{4}\cdot\frac{(x+y)^4}{(1+\theta x+\theta y)^4}, \qquad (0<\theta<1).$$